Name_____

Set up the integral and calculate the volume "exactly" (accurate to 4 decimal places) using your newly acquired calculus prowess. Use a computer graphing utility such GraphCalc (Windows –www.graphcalc.com) or Grapher (Mac OS X – Math Lab) to determine the 2D region and any intersection points.

1. $f(x) = 3 - x \quad g(x) = \dfrac{x^2}{\sqrt{x^3+3}} \quad and \quad x = 0$
   a. Rotate about the x-axis
   b. Rotate about the line y = -1
   c. Rotate about the line y = 4
   d. Cross sections perpendicular to the x-axis that are rectangles with height 3 times the base

2. $f(x) = 6 - \dfrac{x}{2} \quad g(x) = \dfrac{0.8(x-1)^2}{\sqrt{x^3+3}} \quad and \quad x = 0$
   a. Cross sections perpendicular to the x-axis that are semi-circles.
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are equilateral triangles

3. $f(x) = 3 - x \quad g(x) = \dfrac{x^2}{\sqrt{x^3+3}} \quad and \quad \mathrm{y} = 0$
   a. Rotate about the x-axis
   b. Rotate about the line y = -1
   c. Rotate about the line y = 2

4. $f(x) = 6 - \dfrac{x}{2} \quad g(x) = \dfrac{3(x-1)^2}{\sqrt{x^3+3}} \quad and \quad \mathrm{y} = 0$
   a. Cross sections perpendicular to the x-axis that are triangles with height equal to the base
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are semi-circles.

5. $f(x) = -2x^2 + 6\mathrm{x} + 1 \quad g(x) = x + 0.2x^3 \quad and \quad \mathrm{x} = 0$
   a. Rotate about the x-axis
   b. Rotate about the line y = -2
   c. Rotate about the line x = 3

6. $f(x) = -0.7x^2 + 5\mathrm{x} + 1 \quad g(x) = 0.1x + 0.03x^3 \quad and \quad \mathrm{x} = 0$
   a. Cross sections perpendicular to the x-axis that are semi-circles.
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are triangles

7. $f(x) = 1 - e^{-x} \quad g(x) = \ln|x| \quad and \quad \mathrm{x} = 0$
   a. Rotate about the y-axis
   b. Rotate about the line x = -1
   c. Rotate about the line y = 2

8. $f(x) = 5 - 5e^{-x}$ $\quad g(x) = 2.3\ln|x-1|$
   a. Cross sections perpendicular to the x-axis that are equilateral triangles
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are semi-circles

9. $f(x) = 4e^{-0.3x}$ $\quad g(x) = \ln\left|\frac{x}{2}\right|$ *and* $x = 0$
   a. Rotate about the y-axis
   b. Rotate about the vertical line passing through the intersection of the 2 functions in the 1st quadrant.
   c. Cross sections perpendicular to the x-axis that are rectangles with height 2x the base.
   d. Rotate about the x-axis

10. $f(x) = 5e^{-0.2x}$ $\quad g(x) = \ln\left|\frac{x}{3}\right|$ *and* $x = 0$
   a. Cross sections perpendicular to the x-axis that are semi-circles.
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are rectangles with height equal to ½ the base.

11. $f(x) = 0.3x^2 + 0.5$ $\quad g(x) = -0.2x^3 + 3x$
   a. Rotate about the line x = -2
   b. Rotate about the line x = 3.5
   c. Rotate about the line y = -1
   d. Cross sections perpendicular to the x-axis that are rectangles with height equal to 2.5 times the base

12. $f(x) = 0.1x^2 + 0.5$ $\quad g(x) = -0.1x^3 + 4x$
   a. Cross sections perpendicular to the x-axis that are equilateral triangles
   b. Cross sections perpendicular to the x-axis that are squares
   c. Cross sections perpendicular to the x-axis that are semi-circles.

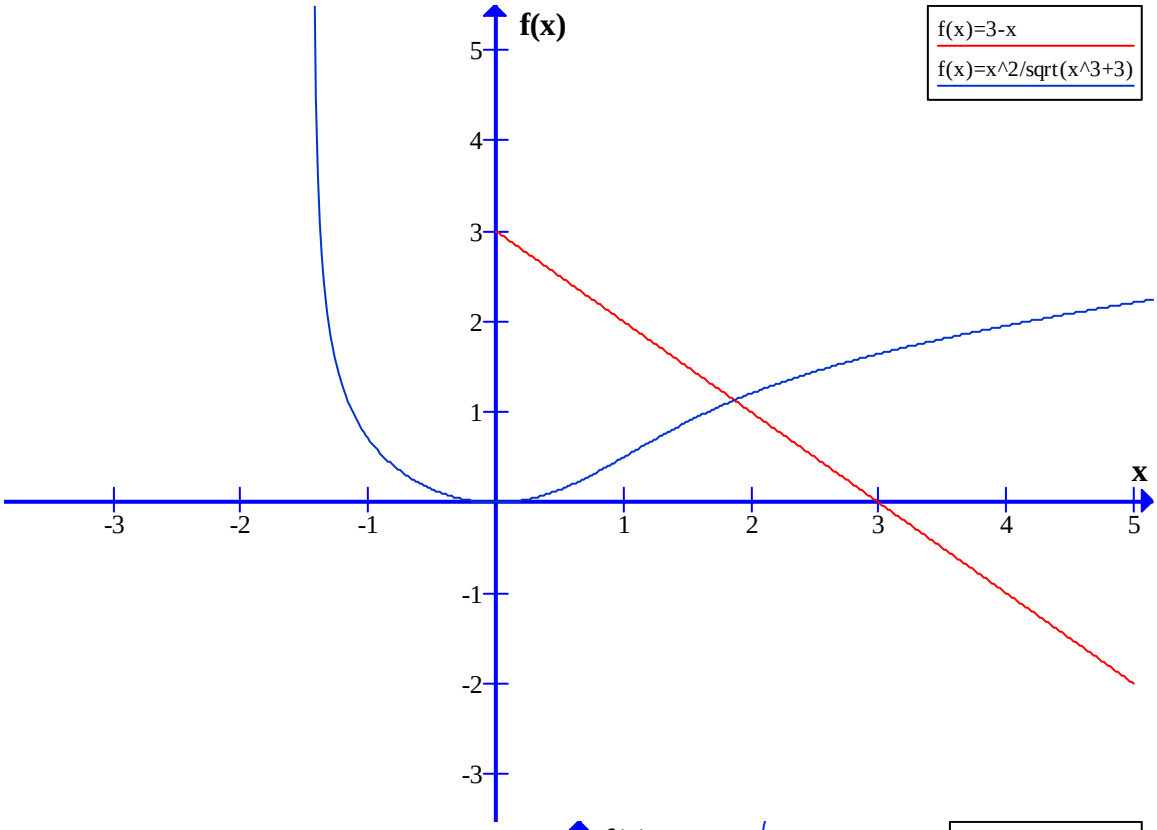
13. $f(x) = (x-2)^3 - 2x + 8$, y=0, x=3.5
   a. Rotate about the x-axis
   b. Rotate about the line y = 5.5
   c. Rotate about the line y = -0.5
   d. Cross sections perpendicular to the x-axis that are rectangles with height equal to 2 times the base.

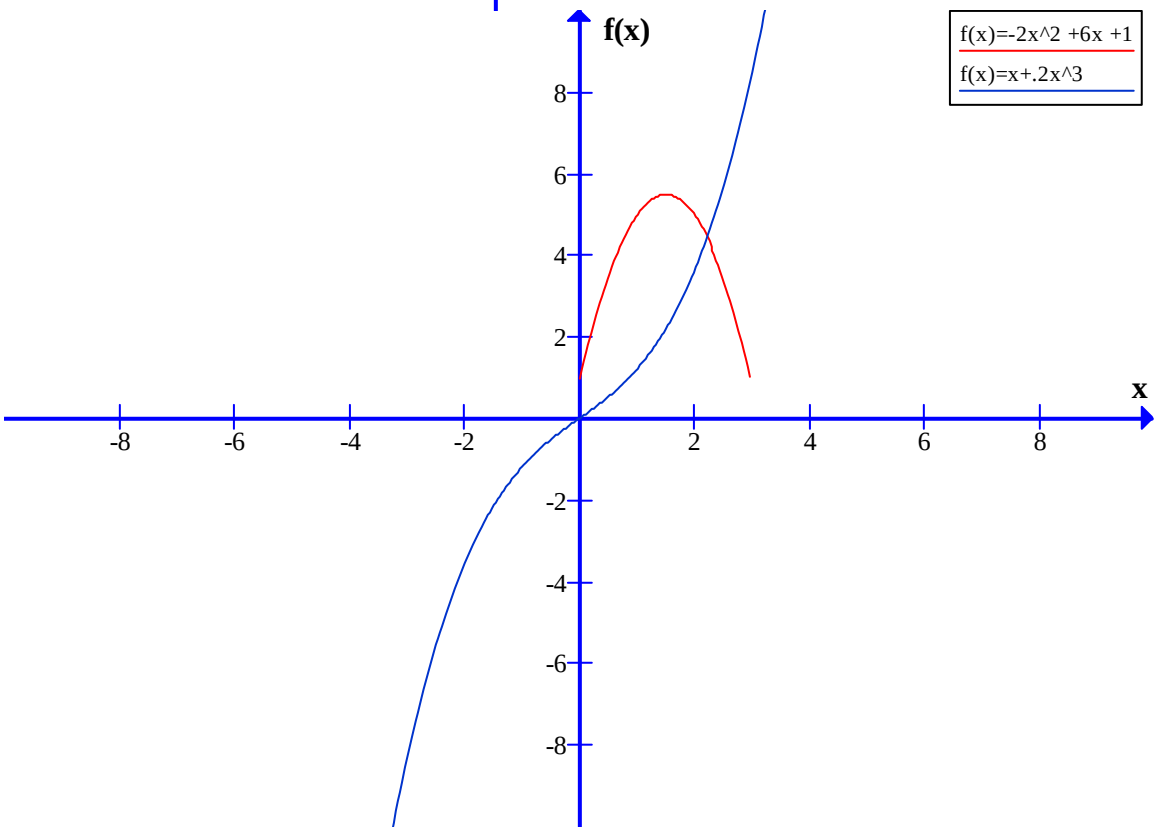
14. $f(x) = (0.9x - 1.5)^3 - 1.5x + 3.5$, $\quad g(x) = (0.3x)^2 + 3$, $\quad x=0$, $x=3.5$
   a. Rotate about the x-axis
   b. Rotate about the line y = 5
   c. Rotate about the line y = -1
   d. Cross sections perpendicular to the x-axis that are isosceles triangles with tall sides equal to 2 times the base.

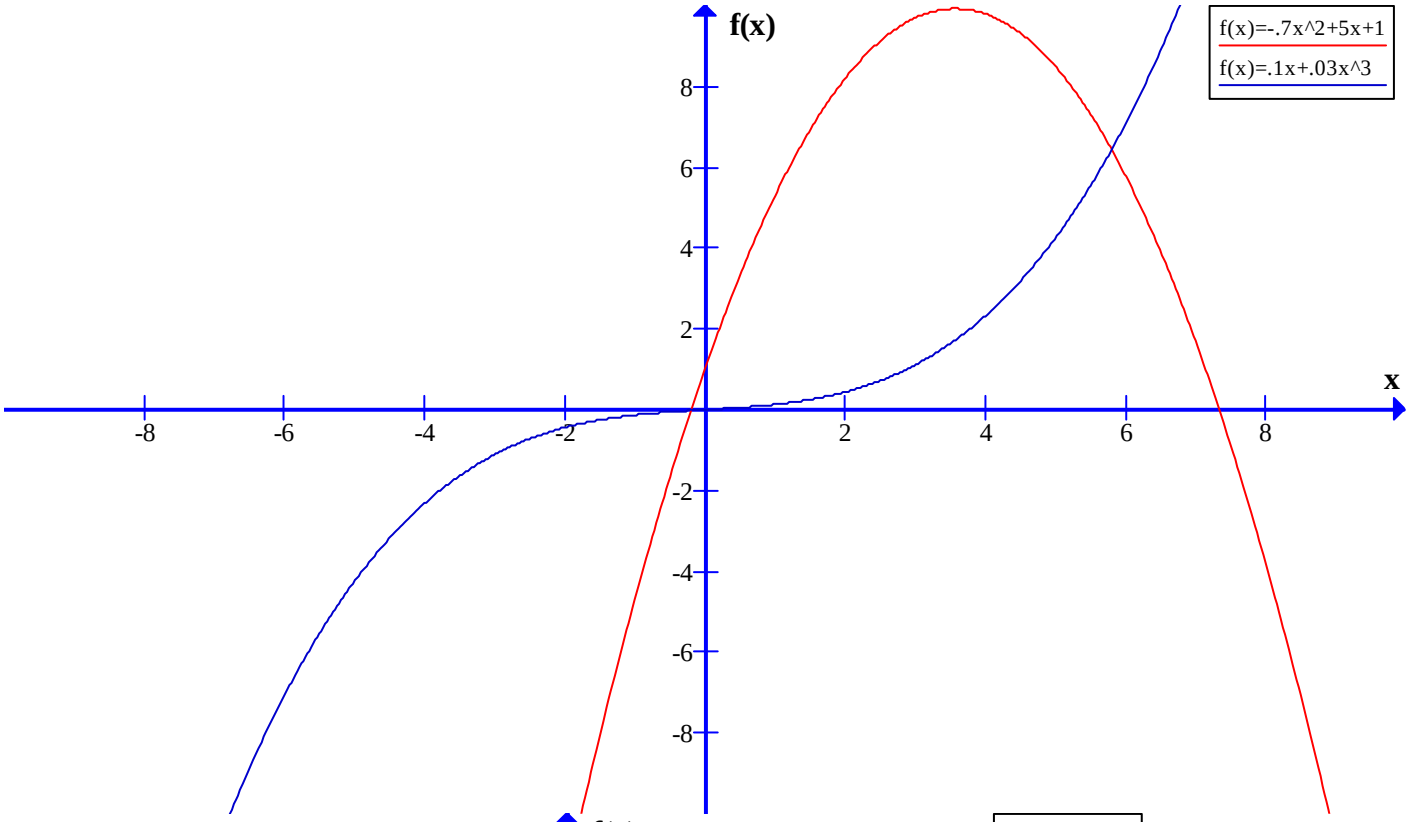
15. $f(x) = (0.9x - 2)^3 - x + 6$, $g(x) = (0.4x)^2$, $x=3.5$
   a. Rotate about the x-axis
   b. Rotate about the line y = 4.5
   c. Rotate about the line y = -1
   d. Cross sections perpendicular to the x-axis that are rectangles with height equal to 2 times the base.

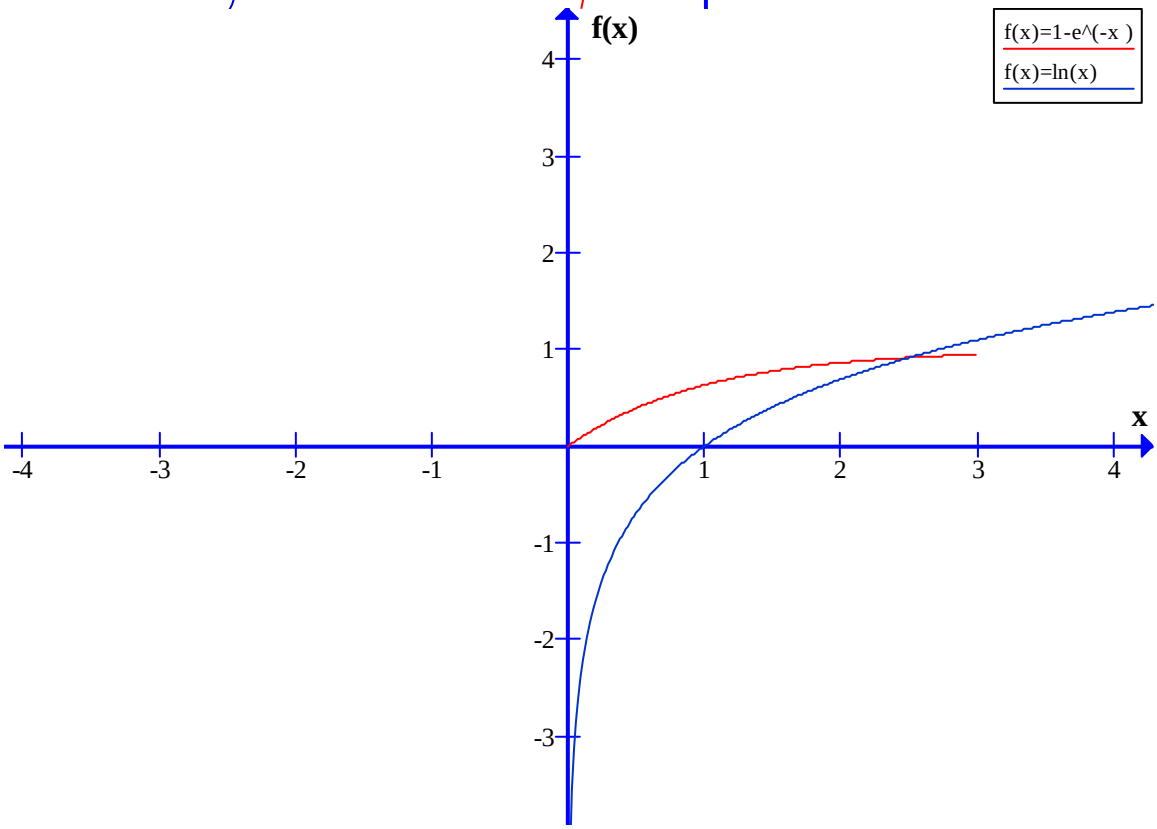
16. $f(x) = (0.5x - 2)^3 - x + 9$, y=0, x=0, x=8
    a. Cross sections perpendicular to the x-axis that are equilateral triangles
    b. Cross sections perpendicular to the x-axis that are squares
    c. Cross sections perpendicular to the x-axis that are semi-circles.
    d. Cross sections perpendicular to the x-axis that are rectangles with height equal to 1/2 times the base.

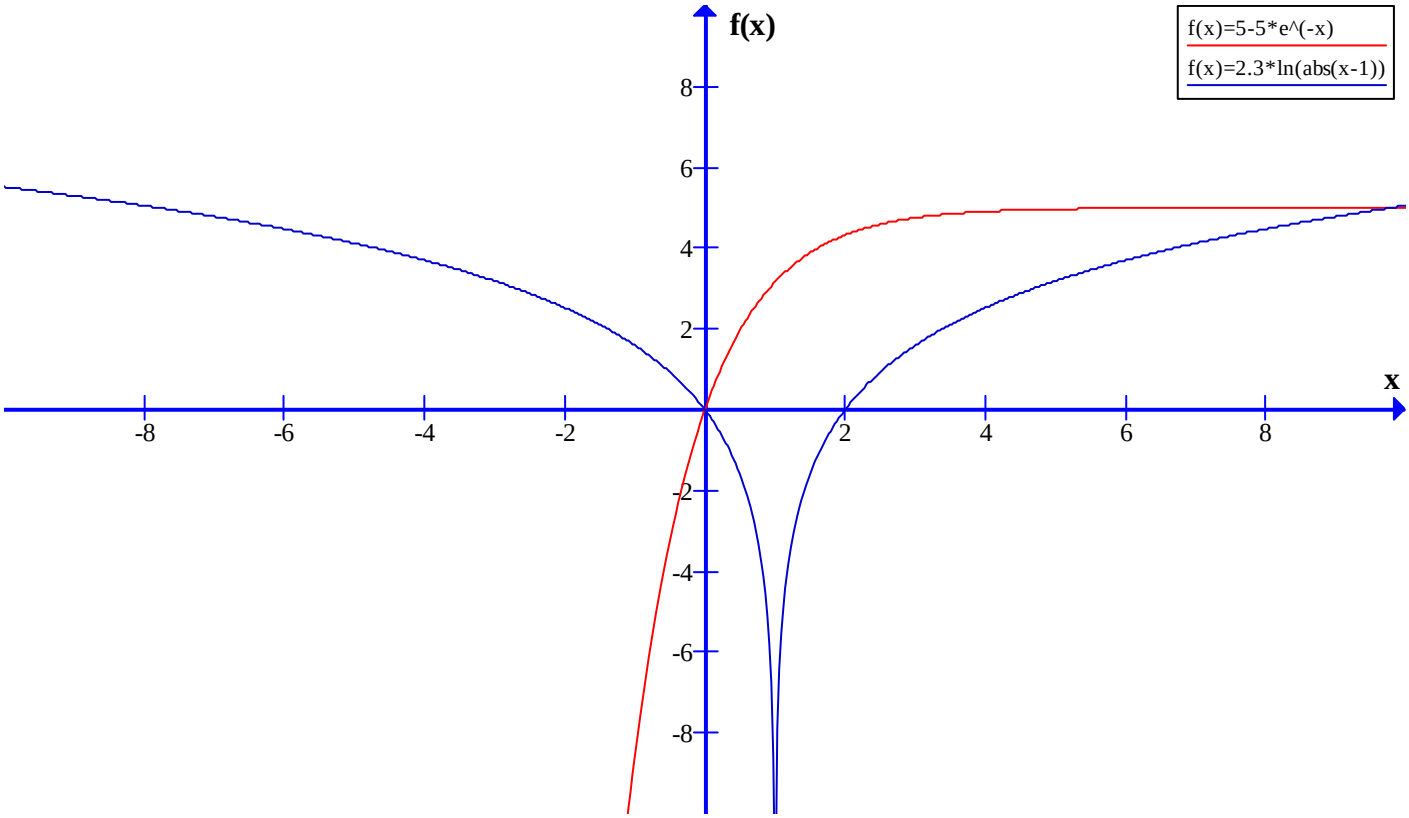
17. $f(x) = (0.5x - 2)^3 - x + 9$, $g(x)=(0.2x)^2$, y=0, x=0, x=8
    a. Cross sections perpendicular to the x-axis that are equilateral triangles
    b. Cross sections perpendicular to the x-axis that are squares
    c. Cross sections perpendicular to the x-axis that are semi-circles.
    d. Cross sections perpendicular to the x-axis that are rectangles with height equal to 1/2 times the base.

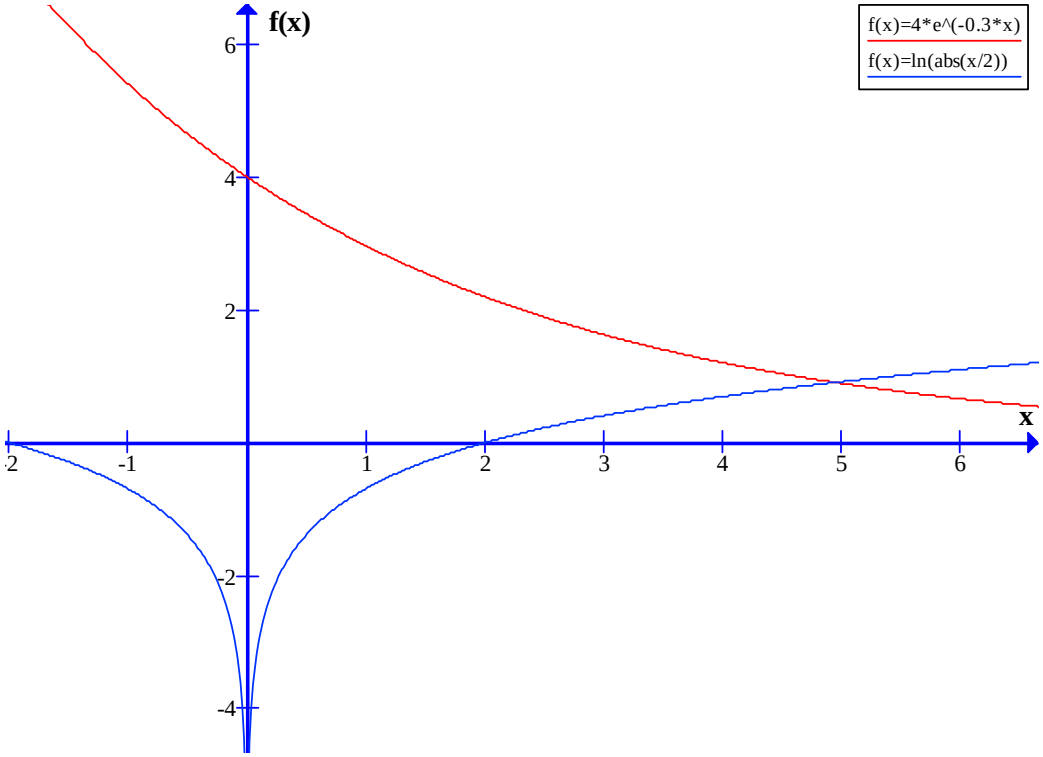

f(x)
x
f(x)=3-x
f(x)=x^2/sqrt(x^3+3)
5
4
3
2
1
-1
-2
-3
-3
-2
-1
1
2
3
4
5

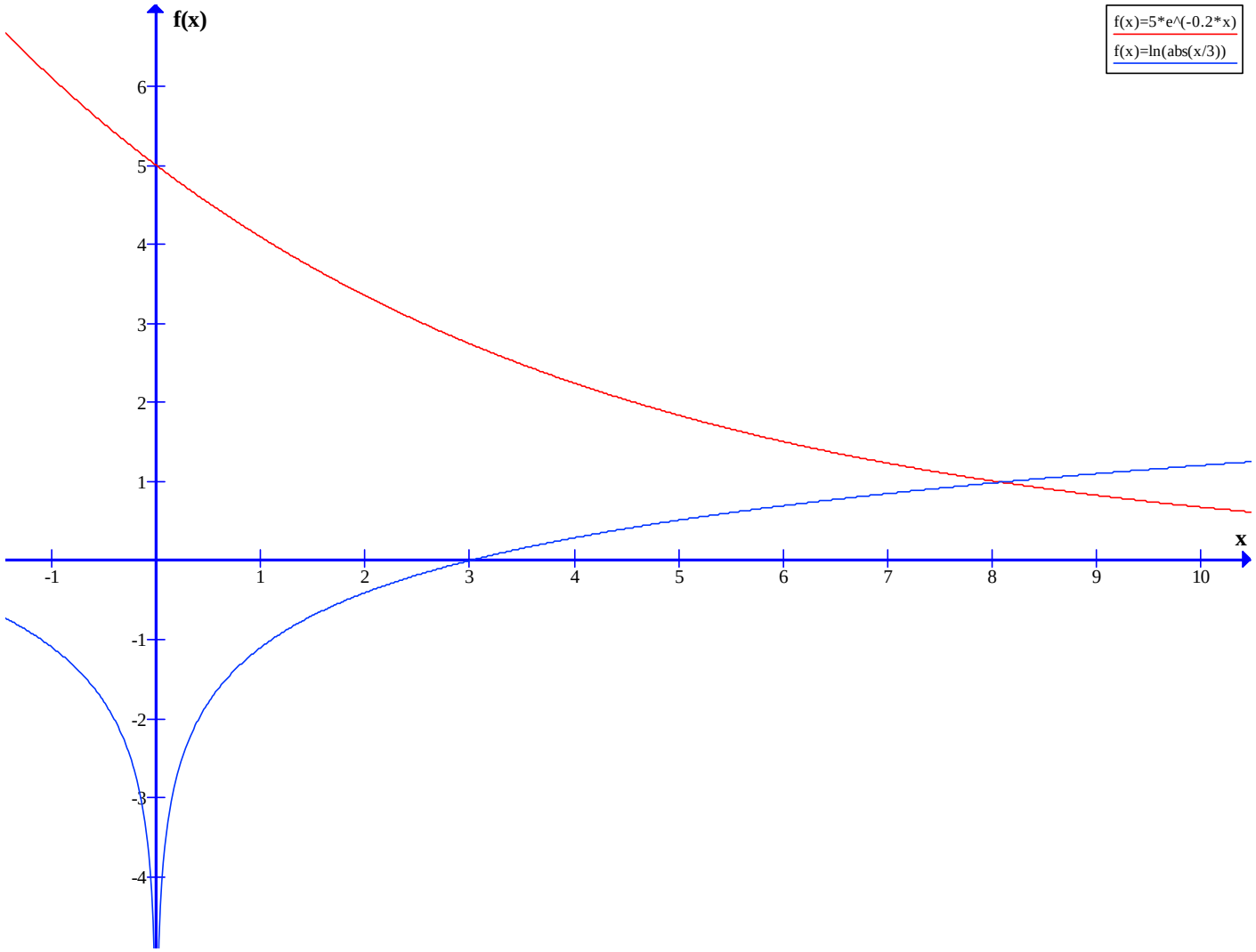

f(x)
x
f(x)=-2x^2 +6x +1
f(x)=x+.2x^3
8
6
4
2
-2
-4
-6
-8
-8
-6
-4
-2
2
4
6
8

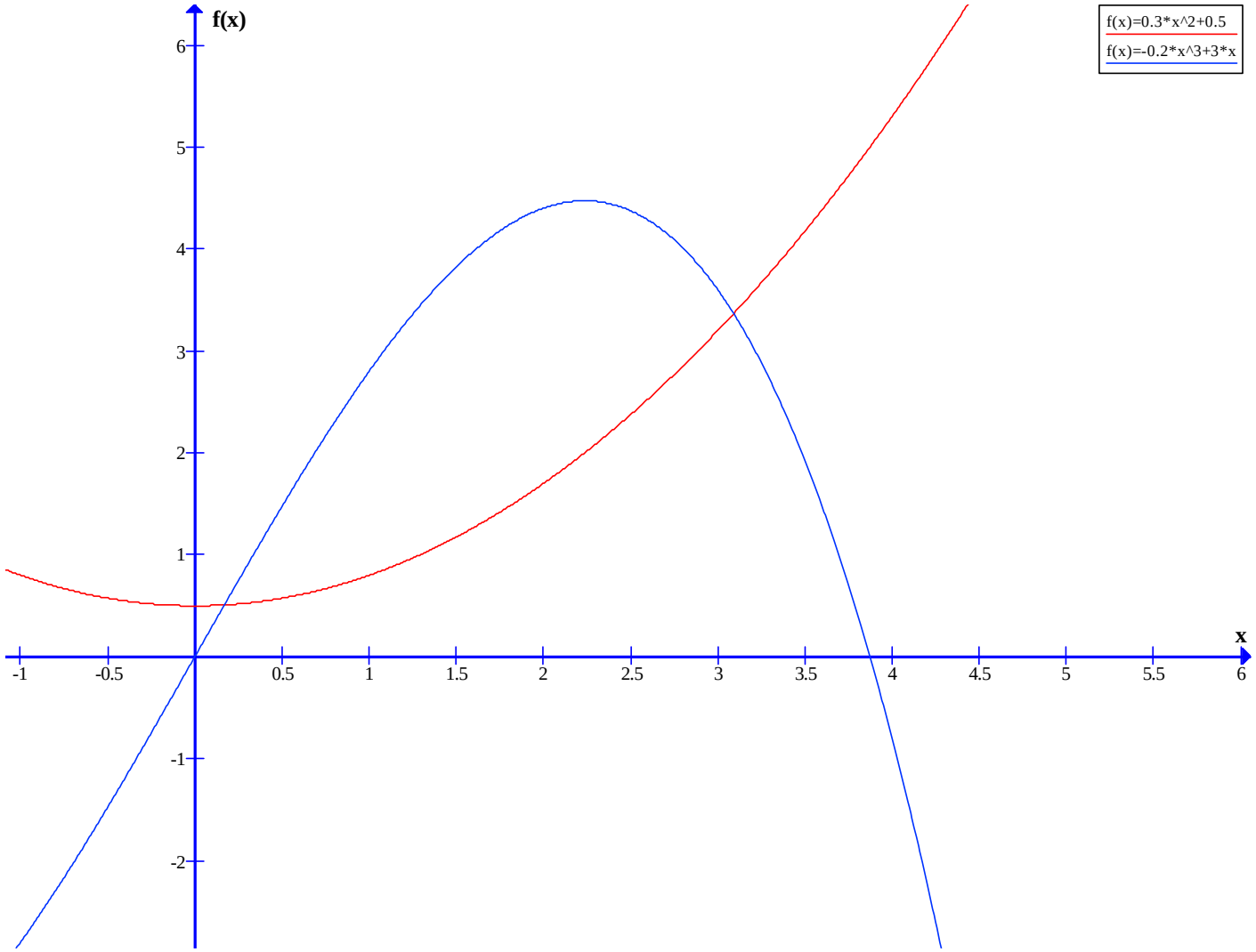

f(x)
x
f(x)=-.7x^2+5x+1
f(x)=.1x+.03x^3
8
6
4
2
-2
-4
-6
-8
-8
-6
-4
-2
2
4
6
8

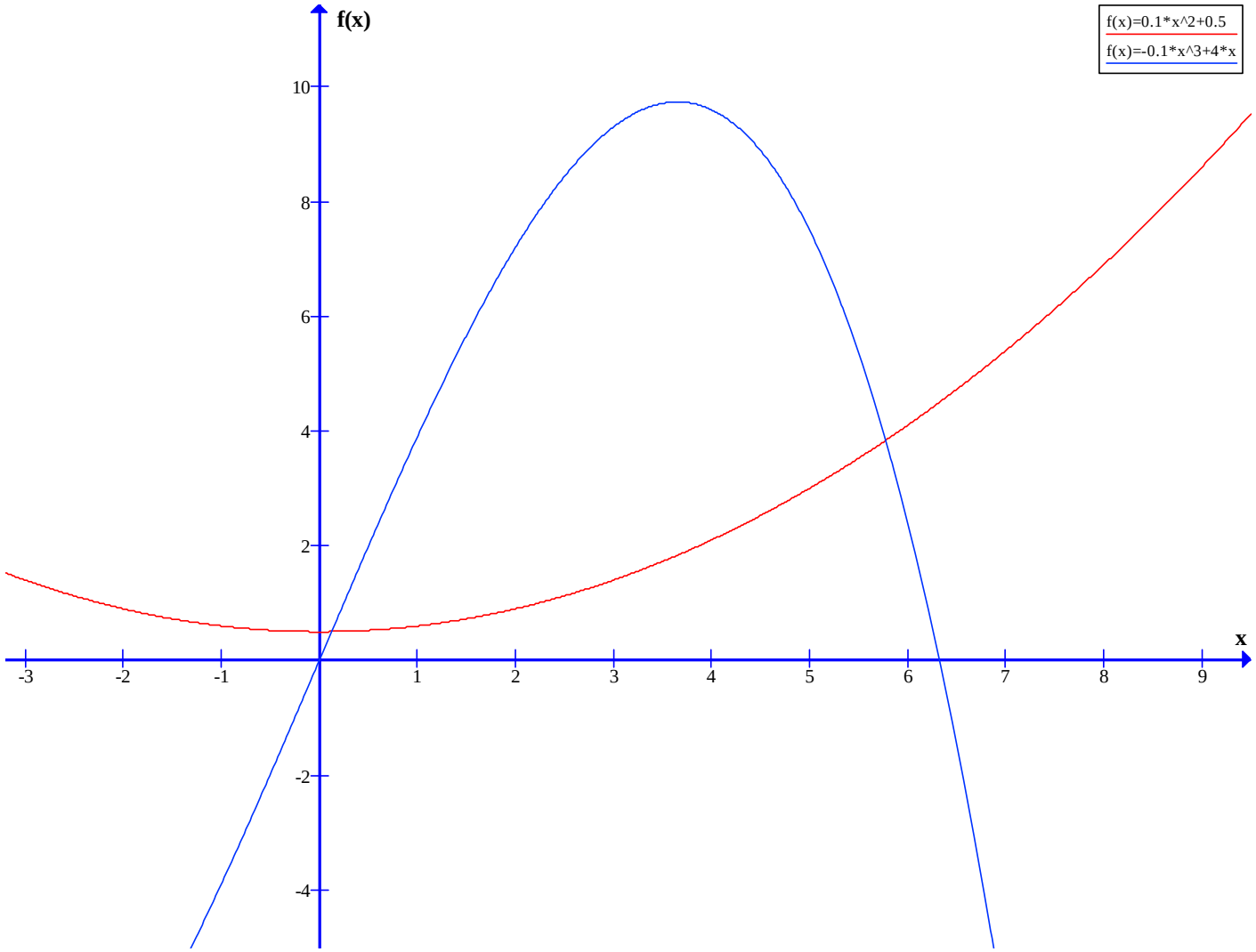

f(x)
x
f(x)=1-e^(-x)
f(x)=ln(x)
4
3
2
1
-1
-2
-3
-4
-3
-2
-1
1
2
3
4


f(x)
f(x)=5-5*e^(-x)
f(x)=2.3*ln(abs(x-1))
x
8
6
4
2
-2
-4
-6
-8
-8
-6
-4
-2
2
4
6
8


f(x)
f(x)=4*e^(-0.3*x)
f(x)=ln(abs(x/2))
x
6
4
2
-2
-4
-2
-1
1
2
3
4
5
6


f(x)
x
f(x)=5*e^(-0.2*x)
f(x)=ln(abs(x/3))
6
5
4
3
2
1
-1
-2
-3
-4
-1
1
2
3
4
5
6
7
8
9
10


f(x)
x
f(x)=0.3*x^2+0.5
f(x)=-0.2*x^3+3*x
6
5
4
3
2
1
-1
-2
-1
-0.5
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6


f(x)
f(x)=0.1*x^2+0.5
f(x)=-0.1*x^3+4*x
10
8
6
4
2
-2
-4
x
-3
-2
-1
1
2
3
4
5
6
7
8
9